IE-PS 3

1. Consider the IPVM. Suppose there are only two bidders and that for $i = 1, 2$ and $\forall v_i \in [0,1]$, $f(v_i) = 1$.

   (a) Formulate the First-Price Sealed-Bid auction (FPA) as a Bayesian Game. Define a strategy.

   (b) Please identify a symmetric, pure-strategy Bayesian Nash equilibrium of the FPA. Can one write the equilibrium strategies as $b_i(v_i) = a + bv_i$ where a and b are constants?

   (c) Consider a modification of the FPA: the highest bidder still wins the object and pays his/her bid but the other bidder must also pay his/her bid. Thus, both bidders must pay their bids. Please identify a symmetric, pure-strategy Bayesian Nash equilibrium of the modified auction. How do the equilibrium bids differ from those in the FPA? Can one express the equilibrium strategies $b_i(v_i)$ as a polynomial?

   (d) Please demonstrate that the FPA generates higher expected revenue for the seller than the modified auction.

2. Consider the IPVM. Please find a symmetric, pure-strategy Bayesian Nash equilibrium of the following modification of the FPA: The highest bidder still gets the object and pays his/her bid; the seller pays to any bidder submitting a bid of $b \in I\!R_+$ the amount $S(b) = \int_0^b F^{n-1}(x)dx$. How does this auction compare to the Second-Price, Sealed-Bid Auction (SPA) in terms of revenue?

3. Consider the IPVM. The seller chooses a First Price Sealed Bid Auction with an entree fee c (that bidders must pay to participate in the auction) and an (announced) reserve price r.

   (a) Please identify equilibrium strategies.

   (b) Suppose the function $v - \frac{1-F(v)}{f(v)}$ is monotone increasing in v. Please find the combinations (r, c) that maximize the seller's expected revenue.

   (c) Do the pairs of optimal combinations (r, c) depend on the number of potential buyers? Are the optimal combinations different from those found in a Second Price Sealed Bid Auction?